

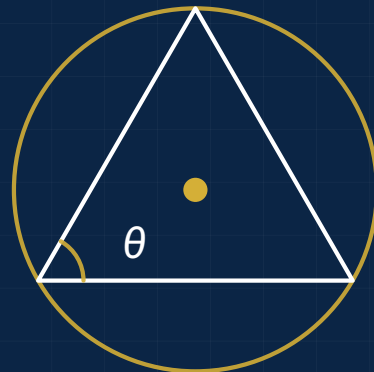


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CHAPTER 4 • GEOMETRY & MEASURES • TOPIC 6

Circles, Arcs & Sectors

A sector is a fraction of a circle: $\frac{\theta}{360}$. Multiply the circle's area or circumference by it — and remember the two radii in a perimeter.



4

VARIANTS

26

PAST-PAPER PARTS

49

PRACTICE QUESTIONS

YOU WILL MASTER

V1 Circle shapes

V2 Arc length & sector area

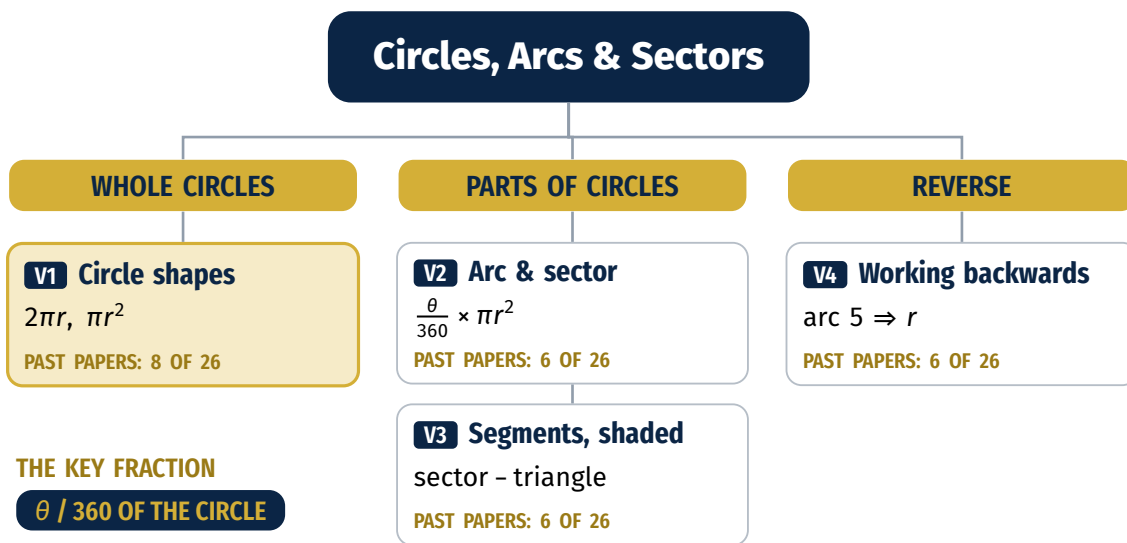
V3 Segments & shaded regions

V4 Working backwards

TOPIC MAP · EVERY VARIANT ON ONE PAGE

Circles at a glance



Your notes


THE KEY FRACTION

 $\theta / 360$ OF THE CIRCLEArc: of $2\pi r$. Sector: of πr^2 .

★ How often does each variant come up?

26 past-paper questions (Jan 2021 → Nov 2025); 49 practice parts — grouped by type.

	PAST PAPERS (OUT OF 26)	PRACTICE PARTS
V1 Circle shapes	8	23
V2 Arc length, sector area	6	17
V3 Segments, shaded regions	6	6
V4 Working backwards	6	3

WHAT THIS TELLS YOU On every paper, often twice: an easy sector area (2 marks, Q11–15) and a harder segment or “find r ” question (4–6 marks, Q20–25). The fraction $\frac{\theta}{360}$ is in every one.

BEFORE YOU START

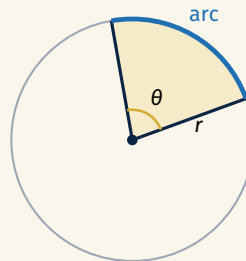
The big idea



Your notes
A sector is a slice of pizza.

An angle of θ at the centre takes $\frac{\theta}{360}$ of the whole circle — of its **area** and of its **circumference**.

The sector's perimeter is the arc **plus two radii**.



The formulas
Circumference

$$2\pi r = \pi d$$

The distance round.

Area

$$\pi r^2$$

Radius, not diameter.

Arc length

$$\frac{\theta}{360} \times 2\pi r$$

Part of the circumference.

Sector area

$$\frac{\theta}{360} \times \pi r^2$$

Part of the area.

Segment
Segment area

$$\text{sector} - \frac{1}{2}r^2 \sin \theta$$

Slice minus the triangle.

Exact answers

$$12\pi + 18$$

"In terms of π ": leave π in.

**CALCULATOR FIRST, THEN CHECK**

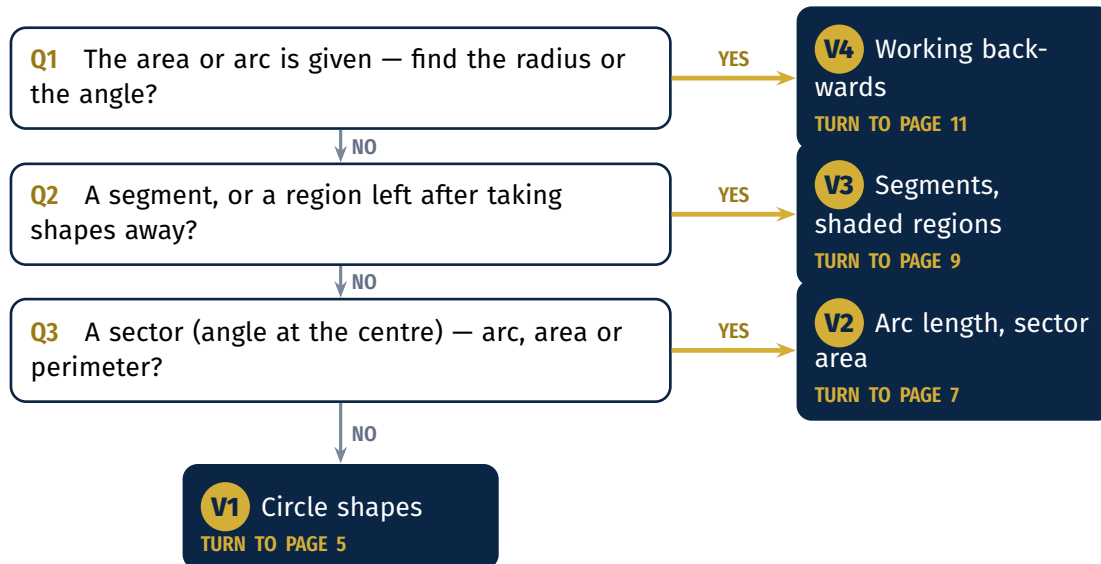
Calculator first, then check: a sector of 90° must be a quarter of πr^2 . Keep full values until the last step.

COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?

 Your notes

Read the question, then walk down the ladder. The first **YES** tells you the variant — and the page to turn to.



COLOUR CODE IN THE WORKING

Blue = letter terms $4x$, $-3y$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of “How do I do it?” each line is.

ON EVERY VARIANT SPREAD

- | | |
|---|---|
|  What is it? |  When you see → you do |
|  How do I do it? |  Watch out |
|  Worked example |  Exam tip |
|  Check yourself |  Calculator check |

V1

VARIANT 1 MOST COMMON IN THE EXAM

Circle Shapes

Radius first; circumference $2\pi r$, area πr^2 ;
halves and quarters, plus the straight
edges.

SEEN IN PAST PAPERS

8 / 26

PRACTICE QUESTIONS: 23


Your notes

Came up in: Jan 2021 P1H Q8 · Jan 2022 P2HR Q9 · Jun 2022 P2H Q20 · May 2023 P1HR Q4 · Nov 2023 P1H Q5 · Jun 2024 P2HR Q5
· Jun 2024 P2H Q21 · Nov 2025 P2H Q6

? What is it?

In plain words

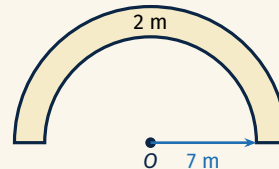
Shapes built from circles, semicircles and straight lines. Find each **radius**, then add the curved parts (πr for a semicircle) and the **straight parts**.

A circle inside a square touching it: diameter = side.

IT LOOKS LIKE

semicircular path

circle in a square



How do I do it?

1 Radii

Every circle part: what is r ?

2 Curves

Whole $2\pi r$, half πr , quarter $\frac{1}{2}\pi r$.

3 Straight edges

Add the sides on the outside.

4 Round

As asked; units.

RADIUS FIRST The most common variant here.

V1 Circle Shapes

VARIANT 1 · SEE IT DONE, THEN TRY



Your notes
 WHEN YOU SEE → YOU DO

“two semicircles ... the path has a width of 2 m”

→ Inner $r = 7$, outer $r = 9$

“a circle inside a square ... the area of the square is 81 cm^2 ”

→ Side 9 = diameter

“Work out the perimeter”

→ Curves + straight edges

**WORKED EXAMPLE** **PAST PAPER**

January 2021 · Paper 1H · Q8 · 3 marks

A path is between two semicircles with centre O : inner radius 7 m, width 2 m, closed by two straight ends. Work out the perimeter of the path (1 d.p.).

STEP 1

$r = 7 \text{ and } r = 9$

◀ inner, outer

STEP 2

$\pi(7) + \pi(9) = 16\pi$

◀ two semicircular arcs

STEP 3

$16\pi + 2 + 2$

◀ two straight ends

STEP 4

$= 54.27 \dots$

◀

ANSWER 54.3 m✓ $16\pi \approx 50.3$, +4 ✓**WATCH OUT**

$\times 2\pi(7) + 2\pi(9)$

✓ $\pi(7) + \pi(9)$

Each arc is a semicircle: half the circumference.

**EXAM TIP**

List every edge of the perimeter before adding — the straight ends are easy to forget.

**Check yourself**

calculator first · answers on page 13

1.1

Dr Eslam

A circle has radius 5 cm. Circumference and area in terms of π ?

Answer:

1.2

Jun 2024 P2HR

A circle inside a square of area 81 cm^2 , touching all sides. Total length of wire for both?

Answer:

1.3

Nov 2023 P1H

Three semicircles of diameter 20 cm make the shape (perimeter $30\pi + 40$). Perimeter to the nearest cm?

Answer:

V2

VARIANT 2

Arc Length & Sector Area $\frac{\theta}{360}$ of $2\pi r$ or of πr^2 ; perimeter = arc + $2r$.

SEEN IN PAST PAPERS

6 / 26

PRACTICE QUESTIONS: 17



Your notes

Came up in: Jan 2021 P2HR Q11 · May 2021 P1H Q12 · Jan 2022 P1HR Q11 · Jun 2022 P2HR Q13 · Jun 2025 P1H Q11 · Nov 2025 P1H Q15

? What is it?**In plain words**

Find the fraction $\frac{\theta}{360}$, then take that fraction of the whole circle.

A **major** sector (the big slice) uses $360 - \theta$. Perimeter adds the **two radii**.

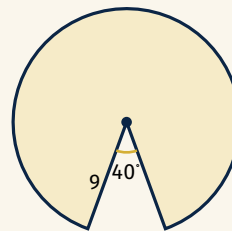
IT LOOKS LIKE

$r = 9$

40° removed

$r = 16$

140°



$$\frac{320}{360} \times 2\pi(9) = 16\pi$$

How do I do it?**1 Angle**The sector's own angle ($360 - \theta$ for the big one).**2 Fraction**

$\frac{\theta}{360}$

3 MultiplyBy $2\pi r$ (arc) or πr^2 (area).**4 Perimeter**Arc + r + r .

$\theta/360 \times \text{WHOLE}$ Then + $2r$ for a perimeter.

V2 Arc Length & Sector Area

VARIANT 2 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

"Calculate the area of sector OPQ" → $\frac{\theta}{360} \pi r^2$

"Work out the length of the arc" → $\frac{\theta}{360} 2\pi r$

"perimeter of the shaded sector" → **Arc + two radii**

**WORKED EXAMPLE****PAST PAPER**

January 2021 • Paper 2HR • Q11 • 4 marks

A circle, centre O , radius 9 cm. A 40° sector is removed. Calculate the perimeter of the shaded (remaining) sector, to 3 s.f.

STEP 1 $360 - 40 = 320^\circ$ ◀ shaded angle

STEP 2 $\frac{320}{360} \times 2\pi \times 9 = 16\pi$ ◀ arc length

STEP 3 $16\pi + 9 + 9$ ◀ two radii

STEP 4 $= 68.27 \dots$ ◀

ANSWER 68.3 cm✓ arc ≈ 50.3 cm, nearly the whole circumference ✓**WATCH OUT**✗ using 40° ✓ 320°

The shaded sector is the big one.

**EXAM TIP**

Write the fraction ($\frac{320}{360}$) on its own line first.

**Check yourself**

calculator first • answers on page 13

2.1

Jun 2025 P1H

Sector OABC: radius 16 cm, angle 140° . Area (3 s.f.)?

Answer:

2.2

May 2021 P1H

A sector of radius 7 cm and angle 50° . Arc length (1 d.p.)?

Answer:

2.3

Jun 2022 P2HR

Sector AOB: $OA = OB = 5.2$ cm, angle 67° . Perimeter (3 s.f.)?

Answer:

V3

VARIANT 3

Segments & Shaded Regions

Segment = sector – triangle ($\frac{1}{2}r^2 \sin \theta$).
 Shaded = big shape – the rest.

SEEN IN PAST PAPERS

6 / 26

PRACTICE QUESTIONS: 6


 Your notes

Came up in: Jan 2021 P2HR Q20 · May 2021 P1H Q25 · May 2023 P1H Q22 · Nov 2023 P2H Q20 · Nov 2024 P2H Q25 · Jun 2025 P1HR Q25

? What is it?

In plain words

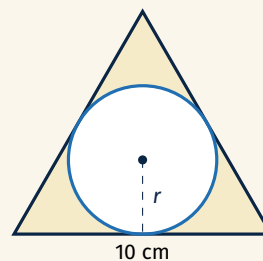
A **segment** is a sector with the triangle cut off: sector $-\frac{1}{2}r^2 \sin \theta$.

Shaded regions: area of the **outer shape** minus what is **not** shaded. A circle inside an equilateral triangle has radius $\frac{\text{side}}{2\sqrt{3}}$.

IT LOOKS LIKE

segment 200 cm²

incircle of a triangle



How do I do it?

1 Outer

Area of the shape around (triangle, sector, square).

2 Inner

Area of what is removed (circle, triangle).

3 Subtract

Shaded = outer – inner.

4 Round

Keep exact values until the end.

OUTER – INNER Segment = sector – triangle.

V3 Segments & Shaded Regions

VARIANT 3 · SEE IT DONE, THEN TRY

 Your notes
WHEN YOU SEE → YOU DO

“the shaded segment” → **Sector** $-\frac{1}{2}r^2 \sin \theta$

“A circle is drawn inside the triangle ... tangents” → **Inradius:** $\tan 30^\circ \times \frac{\text{side}}{2}$

WORKED EXAMPLE **PAST PAPER**

November 2023 · Paper 2H · Q20 · 4 marks

Equilateral triangle, sides 10 cm, with a circle inside touching all three sides. Find the shaded area (triangle outside the circle), 3 s.f.

STEP 1 $\frac{1}{2} \times 10 \times 10 \sin 60^\circ = 25\sqrt{3}$ ◀ triangle

STEP 2 $r = 5 \tan 30^\circ = 2.887$ ◀ radius (from the midpoint of a side)

STEP 3+4 $\pi r^2 = \frac{25\pi}{3} = 26.18$; $25\sqrt{3} - 26.18 = 17.12$ ◀ circle; subtract

ANSWER 17.1 cm²

✓ circle is about 60% of the triangle ✓

WATCH OUT✗ $r = 5$ (half the side)✓ $r = 5 \tan 30^\circ$

The circle touches the sides; its centre is a third of the way up.

EXAM TIP

Name each piece's area on a line of its own before subtracting.

Check yourself

calculator first · answers on page 13

3.1

Dr Eslam

A sector has radius 6 cm and angle 90° . Find the area of the segment.

Answer:

3.2

Jan 2021 P2HR

Angle $ABC = 75^\circ$ at the circumference; the segment cut off by AC has area 200 cm^2 . Radius?

Answer:

3.3

Nov 2024 P2H

A circle of radius x inside an equilateral triangle; shaded area = nx^2 . Find n (3 s.f.).

Answer:

V4

VARIANT 4

Working Backwards

Write the arc or area formula with the unknown, set it equal to the given value, solve.

SEEN IN PAST PAPERS

6 / 26

PRACTICE QUESTIONS: 3


 Your notes

Came up in: Jan 2021 P2H Q22 · Jan 2022 P1H Q16 · Jun 2022 P1H Q19 · Jun 2024 P2HR Q13 · May 2024 P1H Q20 · Nov 2024 P1H Q16

? What is it?**In plain words**

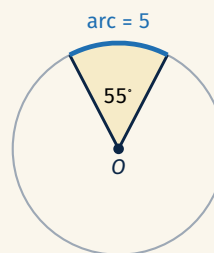
The arc length or area is **given**. Write its formula with the **unknown** (r or θ), set it equal and solve.

Then use the radius for what is asked (area of the circle, perimeter ...).

IT LOOKS LIKE

arc 5 cm

55°; area = 2 × circle



$$\frac{55}{360} \times 2\pi r = 5$$

How do I do it?**1 Formula**Arc or sector formula with r (or θ).**2 Equation**

Equal to the given value.

3 SolveFor r ; square root if r^2 .**4 Use it**

Answer what the question asks.

FORMULA = GIVEN Then use r .

V4 Working Backwards

VARIANT 4 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

"the length of the arc ABC is 5 cm ...
work out the area of the circle" → **Arc** → $r \rightarrow \pi r^2$

"The area of sector S is twice the area
of circle C" → **Equation in r**

"Find a formula for P in terms of r " → **Arc + $2r$, factorise r**

WORKED EXAMPLE PAST PAPER

January 2022 · Paper 1H · Q16 · 4 marks

A, B, C are on a circle, centre O; the arc ABC is 5 cm and angle AOC = 55° .
Work out the area of the circle (1 d.p.).

STEP 1 $\frac{55}{360} \times 2\pi r = 5$ ◀ arc formula

STEP 2 $r = \frac{5 \times 360}{110\pi} = 5.209$ ◀ solve

STEP 3 $\pi r^2 = \pi(5.209)^2$ ◀ area

STEP 4 $= 85.24 \dots$ ◀

ANSWER 85.2 cm²✓ circumference 32.7, and $\frac{55}{360}$ of it is 5 ✓**WATCH OUT**

✗ $\frac{55}{360} \pi r^2 = 5$
✓ $\frac{55}{360} 2\pi r = 5$

An arc is a length: use $2\pi r$.**EXAM TIP**Keep r on the calculator (Ans) — don't round it before squaring.**Check yourself**

calculator first · answers on page 13

4.1 Dr Eslam

A sector of angle 60°
has area 24π cm². Find
the radius.

Answer:

4.2 Jun 2022 P1H

Sector S: radius $(r + 7)$,
angle 45° ; circle C:
radius $(r - 2)$. Area of S
 $= 2 \times$ area of C. Find r .

Answer:

4.3 Jun 2024 P2HR

Sector OAB: radius r ,
angle 60° , perimeter P.
Write $P = r(c\pi + k)$.

Answer:

CHECK YOURSELF

Answers



Every answer below was checked with sympy (check_answers.py).

1.1 10π cm, 25π cm²

1.2 64.3 cm

1.3 134 cm

2.1 313 cm²

2.2 6.1 cm

2.3 16.5 cm

3.1 10.3 cm²

3.2 13.7 cm

3.3 2.05

4.1 $r = 12$

4.2 $r = 5$

4.3 $P = r\left(\frac{1}{3}\pi + 2\right)$

THE WHOLE TOPIC ON ONE PAGE

Circles cheat sheet



Your notes
V1 Circles 8/26

$2\pi r, \pi r^2$

Radius first; add straight edges.

V2 Sectors 6/26

$\frac{\theta}{360} \times \text{whole}$

Perimeter $+2r$.**V3 Segments 6/26**

sector $- \frac{1}{2}r^2 \sin \theta$

Outer $-$ inner.**V4 Backwards 6/26**

$\frac{55}{360} 2\pi r = 5$

Formula = given; solve.

**TOP 3 LOST MARKS**

1. diameter used as radius $\times$
2. perimeter without the two radii $\times$
3. the small angle for a major sector $\times$

BOTTOM LINE $\frac{\theta}{360}$ of the circle $\cdot$ arc $+2r \cdot$ segment = sector $-$ triangle $\cdot$ keep r unrounded.